Irradiance + Sunshade Modeling

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PLANETARY SUNSHADE FOUNDATION

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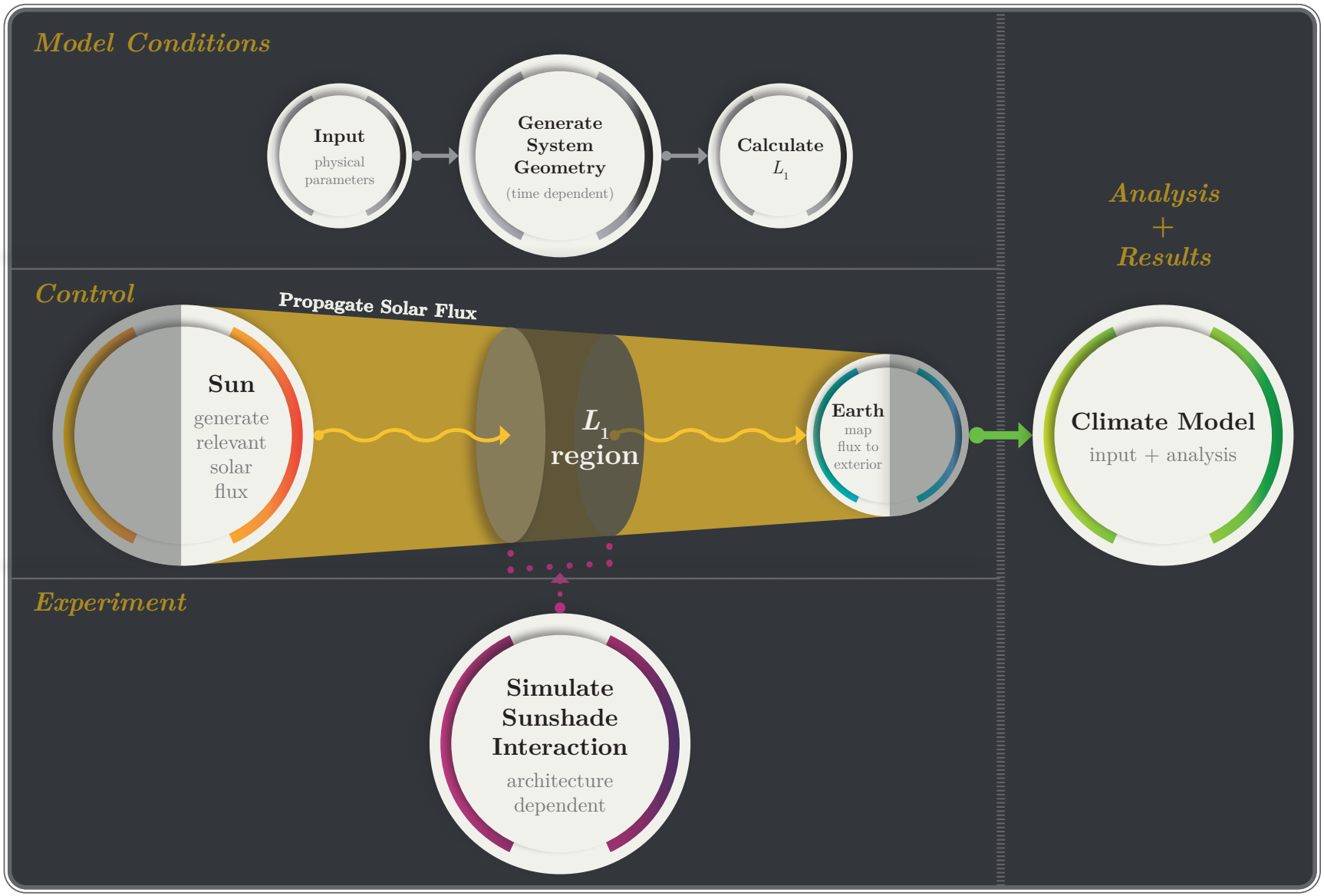

Figure 1.1.1: PFL Model Overview

1 Model Principles

The overall methodology is very direct, succinct, and independent of the forms considered, which allows the modeling of diverse sets of sunshade architectures and arrangements, spherical deviations, planet impact at various atmospheric levels, etc., as well as the introduction of probabilistic interactions with architectures like dust clouds. We'll establish a very simple and completely formalized mathematical framework, then apply it to approximate both irradiance and shade.

1.1 Mathematical Framework

For the mathematical framework, all we'll need to do is establish a system of notation to handle partitions of surfaces (along with their relevant points and vectors), and define a simple result, whereby further specification of its terms will accommodate each of our applications.

For the notation, let α, β, γ index arbitrary *partitions* of surfaces ($S_\star$), their unit normal vectors ($\mathbf{v}_\star$) at arbitrary points ($p_\star \in S_\star$), and the vector between these points ($\mathbf{v}_{\beta\alpha}$, for example).

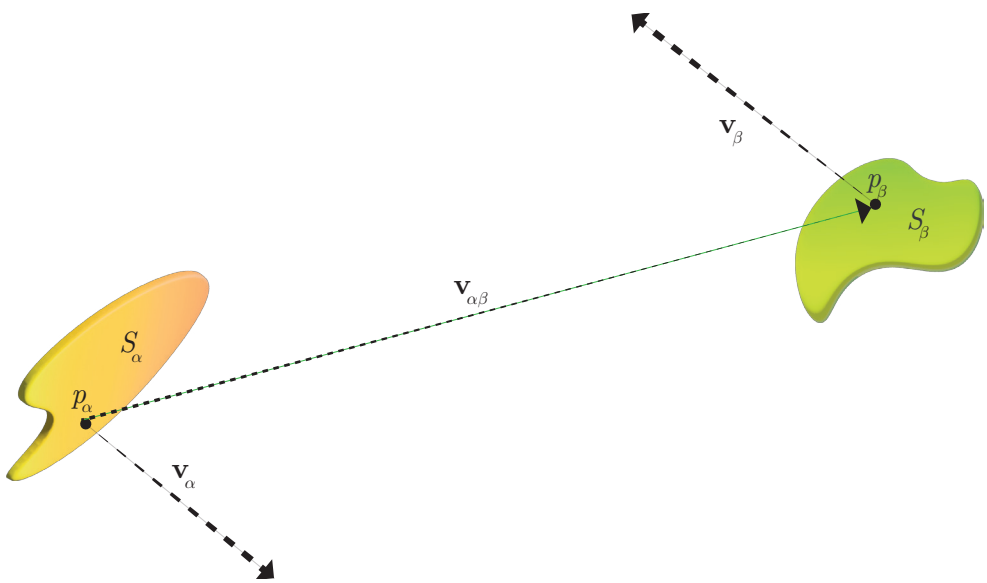

Figure 1.1.1: Exemplifying the notation in the case in which α, β each represent single element partitions of their respective surfaces.

If f denotes some algebraic arrangement functions defined on $S_\alpha \times S_\beta \times S_\gamma$ (which can be generalized to include all the elements above), then[1]

$$\lim_{\max\{|S_\alpha|,\,|S_\beta|,\,|S_\gamma|\} \to 0} \left[\sum_{\star} f\,|S_\star| \right] = \int f\,ds_\star \tag{1.1.2}$$

where $s_\star$ varies over the elements of each partition of $S_\star$.

1 This can be succinctly formalized in a very broad context by simply requiring that our surfaces and their corresponding normal vectors are defined on some *measure space*, where all functions are *measurable* and *bounded* (which corresponds to the physical context needed), whereby the limit in (1.1.2) holds by the *dominated convergence theorem*.

1.2 Approximating Irradiance

To apply our mathematical framework to approximate irradiance at the planet, we'll define the following:

Table 1.2.1: Irradiance Approximation Function Definitions
[via Equation (1.1.2)]

	Definition	**Interpretation**
α, β	star, planet surface partitions	
$\mathcal{L}_{\alpha}^{\alpha\beta}$	$1^{+}(\mathbf{v}_{\alpha} \cdot \mathbf{v}_{\alpha\beta})$	linear transformation[1]: projection of $\lvert S_{\alpha} \rvert$ onto plane normal to $\mathbf{v}_{\alpha\beta}$
φ	$L_{\alpha} \sum a_i (\mathcal{L}_{\alpha}^{\alpha\beta})^i$	luminosity factored by limb darkening[2]
σ	$1/v_{\alpha\beta}^2$	$1/r^2$ scaling (of flux magnitude)
$\mathcal{L}_{\beta}^{\beta\alpha}$	$1^{+}(\mathbf{v}_{\beta} \cdot \mathbf{v}_{\beta\alpha})$	linear transformation: projection of $\lvert S_{\beta} \rvert$ onto plane normal to $\mathbf{v}_{\beta\alpha}$

$$f : \mathcal{L}_{\alpha}^{\alpha\beta} \varphi \, \sigma \mathcal{L}_{\beta}^{\beta\alpha}$$

1 Composed with 1^{+}, the indicator function on the positive reals, assuming zero irradiance contribution when angles $\mathbf{v}_{\alpha} \cdot \mathbf{v}_{\beta\alpha} \notin (-\pi/2, \pi/2)$.
2 The luminosity and limb darkening coefficients are wave-length dependent.

whereby the irradiance at any partitioned surface element (S_{β}) of the planet is then approximated by summing (1.1.2) over α using f as defined above.

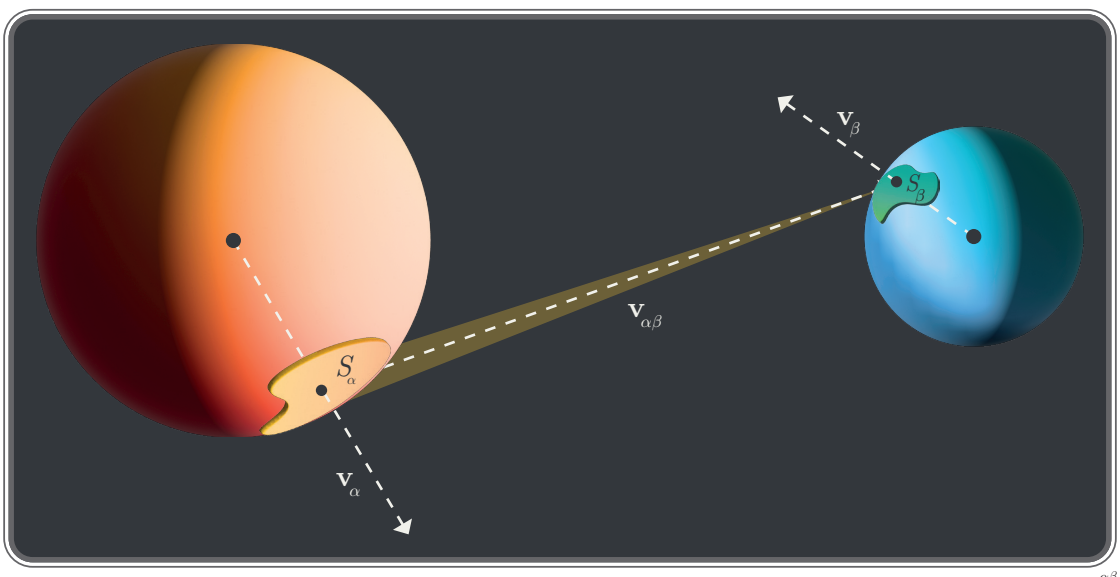

Figure 1.2.2: In words: fixing any two partitioned surface elements (S_{α}, S_{β}) on the star and planet, $\mathcal{L}_{\alpha}^{\alpha\beta}$ projects the surface area $\lvert S_{\alpha} \rvert$ geometrically in the direction of $\mathbf{v}_{\alpha\beta}$, φ adjusts the star's luminosity for limb darkening in this direction, σ accounts for the standard $1/r^2$ scaling, and $\mathcal{L}_{\beta}^{\beta\alpha}$ projects the irradiance onto S_{β}. This is exact for the specific points (p_{α}, p_{β}), and taken as approximate for the rest of $S_{\alpha} \times S_{\beta}$, whereby the result approaches the intuitive ideal of integrating over all points to reproduce the indident cone of light.

At the heart, the model sacrifices a much cleaner Gaussian flux evaluation to gain more detailed propagation information applicable to shading effects. This then relies on a certain precision when describing star luminosity's dependence on angle of incidence (hence *limb darkening*).

Algorithmically, shifting normal vectors in partitions with a certain regularity (such as longitudinal and latitudinal grids) can offer an efficient means of calculating upper and lower bounds on actual values.

1.3 Approximating Shaded Irradiance

To approximate the shaded irradiance of an *arbitrary* set of objects, we'll instead perform a projection of *shaded area onto the star* instead of *irradiance at the planet* by defining:

Table 1.3.1: Shaded Irradiance Approximation Function Definitions
[via Equation (1.1.2)]

	Definition	**Interpretation**
β, γ	planet, shade[1] surface partitions	
$\dot{\alpha}, \mathbf{v}_{\dot{\alpha}}, \mathbf{v}_{\beta\dot{\alpha}}$	extrapolates $\mathbf{v}_{\beta\gamma}$ to star surface	characterizes line of sight through planet (p_β) and shade (p_γ) to star
$\mathcal{L}_{\gamma}^{\beta\dot{\alpha}}$	$1^{+}(\mathbf{v}_{\gamma} \cdot \mathbf{v}_{\beta\dot{\alpha}})$	linear transformation: projection of $\lvert S_{\gamma}\rvert$ onto plane normal to $\mathbf{v}_{\beta\dot{\alpha}}$
σ	$(v_{\beta\dot{\alpha}} / v_{\beta\gamma})^2$	geometric scaling of projected shade area from p_{γ} to star $\mathrm{p}_{\dot{\alpha}}$
$\mathcal{L}_{\beta\dot{\alpha}}^{\dot{\alpha}}$	$1^{+}(\mathbf{v}_{\beta\dot{\alpha}} \cdot \mathbf{v}_{\dot{\alpha}})$	linear transformation: scaled projection of $\lvert S_{\gamma}\rvert$ onto plane normal to $\mathbf{v}_{\dot{\alpha}}$
φ	(flux $S_{\dot{\alpha}} \to S_{\beta}) \cdot \kappa_{\gamma}$	blocked flux[2] with material absorption factor

$$f : \mathcal{L}_{\gamma}^{\beta\dot{\alpha}} \sigma \mathcal{L}_{\beta\dot{\alpha}}^{\dot{\alpha}} \varphi \quad ^{3}$$

1 Only (say) starward oriented shade surfaces.
2 As calculated in Section 1.2.
3 Here φ is calculated in the same way as was done for the flux coming from a pre-partitioned star surface (see table Table 1.2.1), however here there is no partition of the star. By way of example, the projected area of a 20 km circle at L_1 onto the sun will result in a shaded area on order of 10^{-6} of the total sun's surface, which thereby justifies its treatment at an individual unit of some larger partition that's left undefined.

whereby the irradiance blocked from arriving at any partitioned surface element (S_β) on the planet is then approximated by summing (1.1.2) over γ and using f as defined above.

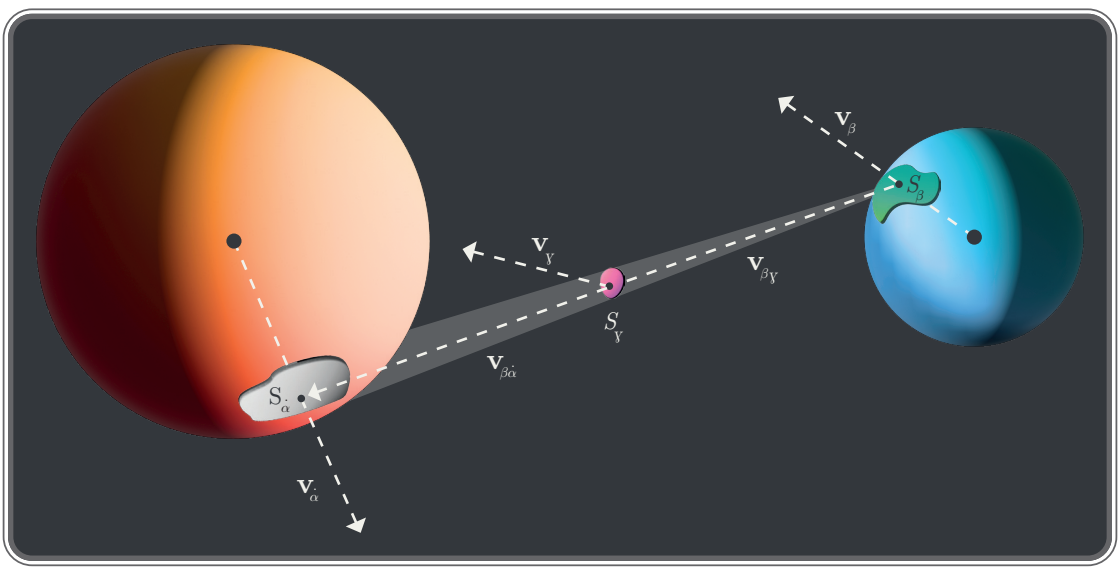

Figure 1.3.2: In words: fixing any two partitioned surface elements (S_β, S_γ) on the planet and the shade, $\mathcal{L}_{\gamma}^{\beta\dot{\alpha}}$ projects the shade's area $\lvert S_{\gamma}\rvert$ geometrically in the direction of $\mathbf{v}_{\beta\dot{\alpha}}$, σ scales this geometrically to the surface of the sun, $\mathcal{L}_{\beta\dot{\alpha}}^{\dot{\alpha}}$ projects the resulting shade to the star surface, φ calculates the irradiation contribution of that shaded surface ($\mathrm{S}_{\dot{\alpha}}$) to S_β, multiplied by a material factor indicating how much was blocked. This is exact for the specific points (p_β, p_γ), and taken as approximate for the rest of $S_\beta \times S_\gamma$, whereby the result approaches the intuitive ideal of integrating over all points to reproduce the shaded cone of light.

As γ was treated as a partition of a *set of arbitrary surfaces*, the model allows complex and multi-bodied sunshade architectures. The potential to overlap can be treated by parametrizing broader (for efficiency) shade surfaces that would naturally accompany the shade's partition. Dust cloud architectures can also potentially be modeled through incorporation of probabilistic models

into κ_γ.

1.4 Limb Darkening And Its Application To Luminosity

Limb darkening describes the observation that the center of a star appears brighter at its center than at its edges. If the observed intensity $I(\theta)$ is assumed to only depend on the angle of incidence $\theta \in (0,\pi/2]$ at which the observation is made with respect to the star's surface, then without loss of generality it can be expressed empirically as a power function.[1]

$$I(\theta) = I(0) \sum a_i \cos^i \theta \tag{1.4.1}$$

The physical requirement for the range to lie in [0,1] then constrains the a_i to sum to unity, whereby the series convergences on the closed interval $\theta \in [0,\pi/2]$.

Multiplying both sides of (1.4.1) by the magnitude of a partitioned surface area of the star then gives luminosity as a function of angle. If the total luminosity of the partitioned surface is known, this can then be used in lieu of $I(0)$ provided that the sum is then normalized by integrating over $\theta \in [-\pi/2, \pi/2]$.[2]

1 Composed with the bijection $\theta \to \cos\theta$. This can also be perceived as the symmetric part of a Fourier series expansion defined over $(0,\pi)$,

2 Normalizing over the full range of incident angles ensures that the surface would emit more than its total luminosity. Change of coordinates isn't necessary, as θ can be interpreted as indicating an arbitrary emission direction, whereby discrete sums (and the limits thereof) should approach the total.

2 Program Overview

2.1 Getting Started

Here we'll start by downloading the program, making a few quick modifications so that Matlab can find everything on your computer, and then running a quick test to ensure everything works.

a) Download

Download the contents of the DropBox Folder (password: *Planetary Sunshade Foundation*) and put them somewhere nice. This should contain three folders: {matlab}, {excel}, and {numerical control}, as well as the current version of the documentation.

b) Tell Matlab where your downloaded programs are located

In the Matlab *Home Tab*:

- Click *Set Path*
- From the *Add Folder* dropdown at the top left click *Add Folder With Subfolders*.
- Identify the path to the downloaded folder {matlab | sunshade models | psf | current version}.

c) Tell the programs where your excel and numerical controls folders are

- Open the program `location_Excel_Folder___0___St` {matlab | sunshade models | psf | current version | file locations}.
- Change the string to the location of the downloaded folder {excel | psf model} on your computer.
- Save and close the program.

```
location_Excel_Folder = 'C:\Users\...\excel\psf model\'
```

Figure 2.1.1: An example of entering the location of the downloaded folder {excel | psf model} as a string into the program `location_Excel_Folder___0___St`.

- Repeat these steps to update the location of your downloaded folder {numerical control} within the program `location_NC_Folder___0___St` {matlab | sunshade models | psf | current version | file locations}.

d) Take a test run in Matlab

Typing the function name `analysis_General___0___X` into the Matlab command window and hitting enter should then output an 11×11 matrix (longitudinal and latitudinal grid, as if you were looking at the surface of the planet) of the solar irradiance incident on each of these partitioned planet subsurfaces.

2.2 Adjusting Analysis Parameters: Parameters In The Code

Analyses parameters can be configured manually or, in the case of outputting an NC file, can be set to a preconfiguration. In analyzing shading effects, the results are both quicker and easier to interpret with a manual configuration, after which point a full evaluation can be performed with an output NC file.

All analysis parameters are either input into the `program input_Model_Parameters___0___Sr` {matlab | sunshade models | psf | current version | input} or the Excel files (shade locations, limb darkening constants). Here we'll go over the program parameters defined in the code, and the next section will cover the Excel files.

a) Analysis + Configuration Types

- `analysis_type` {`'irradiance'`, `'shaded irradiance'`, `'shading factor'`, `'irradiance factor'`} determines what is actually being calculated. *Irradiance* and *shaded irradiance* give the total irradiance (in W/m^2) incident upon, or the total irradiance blocked, for each partitioned subsection of the planet. *Shading factor* then gives the ratio (*shaded irradiance*/*irradiance*), while irradiance factor gives 1 minus this value.
- `configuration` {`'manual'`, `'preconfigured: low resolution'`} determines if the time and partition parameters are determined manually (recommended for evaluation purposes) or are preconfigured to correspond with NC file output. will automatically perform an *irradiance factor* analysis and output the results as an NC file, with date and time stamp, in the folder { numerical control | exports }.

```
model_parameters.analysis_type = 'shading factor';
%-------------------------------------------------------------------------
model_parameters.configuration = 'manual';
```

Figure 2.2.1: Selecting the analysis and configuration type. For a manual analysis, *shading factor* results are easier to interpret than *irradiance factor*, as Matlab will not display enough decimal in its standard output format (which is toggled by the commands `format long` or `format default`).

- `output.type` {`'matlab'`, `'nc file'`} currently has no effect, and anticipates the incorporation of a Matlab preview of results that would otherwise be too large to interpret directly from the NC file output (conversions to yearly averages, for example).

b) Partition Properties

- `phi_interval_S` and `n_phi_S` give the latitudinal range and the number of latitudinal divisions with which to partition the surface of the star. The corresponding values for `theta` then do the same for longitude, and those suffixed with `P` just below do the same for the planet.

  Zero marks dead center along the star planet axis in both cases. The program will then choose representative surface positions and normal vectors in the center of each demarked region. With interval symmetrical about zero, setting both `n_phi` and `n_theta` to 1 will model the body as single plane region normal to the star planet axis.

```
phi_interval_S   = [-90,90];
theta_interval_S = [-90,90];

n_phi_S          = 21;
n_theta_S        = 21;

%-------------------------------------------------------------------------

phi_interval_P   = [-90,90];
theta_interval_P = [-90,90];

n_phi_P          = 11;
n_theta_P        = 11;
```

Figure 2.2.2: Defining star and planet partitions in terms of latitudinal and longitudinal angles and numbers of divisions. A 21×21 partition of the star can be rather accurate as results have a tendance to converge rapidly. An 11×11 partition of the planet generally allows the entire array to be viewed in the Matlab command window. Both afford an efficient means to quickly evaluate the quality of a shading arrangement.

c) Time

Start times, as well as the frequency and number of analysis can all be set manually. Matlab will output the re-

sults for each time as a two-dimensional slice of a higher dimensional matrix: if no spectral information is added for the star, these will then display as separate matrices in the command window.

- `time.start` determines the first day on which the analysis is to be performed. This is to be entered in UTC time, according to the format shown in the image below. Data exists for any date in the years 2020-2025. Any changes to the actual time on that date are ignored downstream, and the program wil automatically evaluate star and planet kinematics at noon.
- `time.frequency` determines a number of days N, such that one analysis is performed every N days.
- `time.periods` determines the total number of analysis to perform at the indicated frequency.

```
model_parameters.time.start     = '2025-01-01T12:00:00';
%-----------------------------------------------------------------------
model_parameters.time.frequency = 180;
%-----------------------------------------------------------------------
model_parameters.time.periods   = 2;
```

Figure 2.2.3: Defining time parameters. `time.periods` indicates the total number of analyses to be performed. So here we have a first analysis performed at `time.start` (January 1, 2025), and another 180 days later. If the initial date preceeds January 1, 2020, or the final date exceeds December 31, 2025, the program will trigger an error as star planet kinematics data is not saved outside of this range.

d) Heliogyro Unit Properties

The properties of an individual heliogyro unit are defined within the code. Each heliogyro is modeled as a polygon through Matlab's `polyshape` function. The program allows for separate definitions of the polygon interior and exterior, and cuts away the interior from the exterior region to simulate the heliogyro's shape.

- `sides.inner, sides.outer` determine the number of sides of the interior and exterior polygon, both of which are regular.
- `radii.inner radii.outer` determine the radii or the interior and exterior polygon (center to vertice).
- `material_irradiance_absorption` can be interpreted in several ways, one of which is giving the fraction of irradiance absorbed by the material, or equivalently the fraction of the total area that will absorb at 100%.

```
model_parameters.shade.heliogyro.sides.inner                    = 52;
model_parameters.shade.heliogyro.sides.outer                    = 52;
%-----------------------------------------------------------------------
model_parameters.shade.heliogyro.radii.inner                    = 10;
model_parameters.shade.heliogyro.radii.outer                    = 20;
%-----------------------------------------------------------------------
model_parameters.shade.heliogyro.material_irradiance_absorption = 0.5;
```

Figure 2.2.4: Defining the properties of the interior and exterior polygon used to model individual heliogyro units. The program then generates both polygons as regular polygons, and cuts out the interior polygon from the exterior polygon to model the heliogyro surface.

2.3 Adjusting Analysis Parameters: Parameters In The Excel Files

Both heliogyro kinematics data and star spectral information are modified within the Excel files.

a) Heliogyro Kinematics Data

- Initial heliogyro positions and orientations (their surface normal vectors) can be entered into the downloaded Excel file { excel | psf | heliogyro kinematics data }.
- The positions and normal vectors for each heliogyro are entered along the different rows.

- Coordinates are relative to a Cartesian coordinate system centered at the current calculated L_1 point, with x corresponding to the star planet axis, y to the tangential motion of the planet, and z normal to the orbital plane.
- Normal vectors don't need to be normalized and there is no preference for positive/negative orientation (as the heliogyros are modeled as surfaces, the program will assign a consistent positive/negative orientation for all normals, as well as normalize the vectors themselves).

a) Star Spectral Data (Limb Darkening)

- Star spectral data (spectral bands, intensities, and limb darkening coefficients) can be entered into the downloaded Excel file { excel | psf | star spectral data }.
- The first row contains default values cited for total intensity. [2]
- Intensities I_v represent the fraction of total intensity attributed to the spectral band.
- Limb darkening coefficients should be entered with respect to the form of equation (1.4.1).
- Any non-attributed intensities will be attributed to the default band.
- The output for each band will be given as a two-dimensional slice of a higher dimensional matrix.

3 Further Details

3.1 L_1 And Basic Line-Of-Sight Calculations

L_1 calculations are currently performed with the simpler equilibrium equation:

$$\frac{Gm_\beta}{v_{\beta\gamma}^2} - \frac{Gm_\alpha}{(v_{\alpha\beta}-v_{\beta\gamma})^2} + \omega_\beta^2(v_{\alpha\beta}-v_{\beta\gamma}) = 0 \tag{3.1.1}$$

and can potentially accomadate radiation pressure on the shades.

$$\frac{Gm_\beta}{v_{\beta\gamma}^2} - \frac{Gm_\alpha}{(v_{\alpha\beta}-v_{\beta\gamma})^2} + \omega_\beta^2(v_{\alpha\beta}-v_{\beta\gamma}) + \frac{2\kappa P_\beta |S_\gamma|}{m_\gamma}\left(\frac{v_{\alpha\beta}}{v_{\alpha\beta}-v_{\beta\gamma}}\right)^2 = 0 \tag{3.1.2}$$

Basic line-of-sight calculations from the planet to the star are performed via the quadratic resulting from (6.3.1).

3.2 Modeling Individual Heliogyros

The interior and exterior contours of heliogyros are modeled as polygons, which is not only more accurate than a circular approach, but reframes an otherwise very difficult problem of intersecting projected circle areas (ellipses, see [1]) into finite vector sets and the well-established domain of wedge products. However, in practice, this has been relagated to Matlab's `polyshape` functions, in combination with matrix manipulation of 3D transformations to provide 2D images within this framework.

3.3 Large Numbers of Heliogyros At L_1

The analysis of large numbers of heliogyros (10^4) touches on computational feasibility, as combinatorically direct methods require analyzing two-unit overlaps on the order of 10^8, (or worse 10^8! for all unit overlaps!), time-dependent positions on the order of (10^3), and partitioned planet surface on the order of 10^5 (high-resolution): i.e. total operations on the order of at least 10^{16}.

Analysis (6.4.1) (and others), or quick counter-examples spacing shades across L_1 depth, show that one point on the planet cannot serve as a reference frame for identifying all possible overlaps. However, extrapolating paths once between all units might be able to provide a complete list of two-unit overlaps for each region in a single calculation.

From there, overlaps between more than two or three units would start to get computationally heavy (as they'd have to be performed at each time frame). This could be met with a probabilistic analysis, as it seems unlikely that many units would actually form a linear arrangement tight enough to overlap from the perspective of any particular region.

4 Results

Table 4.3.1: Irradiance Calculations at Perihilion and Aphelion

	Time (UTC Noon)	**Sun Partition**	**Earth Partition**	**Run Time** (s)	**Calculated I_{max}** (W/m²)	**Expected I_{max}** (W/m²)	**Error**
perihelion	01/01/2025	50×50	51×51	0.86	1335	1311	$+1.8\%$
aphelion	07/04/2025	50×50	51×51	0.90	1427	1417	$+0.71\%$
perihelion	01/01/2025	500×500	11×11	3.77	1335	1311	$+1.8\%$
aphelion	07/04/2025	500×500	11×11	4.12	1427	1417	$+0.71\%$

Calculations used Sun-Earth positions imported from SPICE and cited limb darkening coefficients for total irradiance. Theoretical expected values used 1361 W/m² adjusted for Sun-Earth distances of 1.017 a.u. (perihelion) and 0.983 a.u. (aphelion), respectively. The maximum irradiance value represents the calculation made for the Earth surface partition that is treated as a plane normal to the Sun-Earth axis, which is output by either partitioning the Earth as a single plane or into any odd number of latitudinal and longitudinal grids (whereby the central maximum behaves in the same way). The order of the Earth partition was decreased with the 500×500 Sun partition in order to avoid surpassing Matlab's 7.9 gig matrix limit. Using odd Sun partitions showed no significant difference.

Irradiance calculations are roughly on the order of $|\alpha||\beta|$, while shaded irradiance that of $|\alpha||\gamma|$.

5 References

[1] Hughes and Chraibi. *Calculating Ellipse Overlap Areas.* Computing and Visualization in Science. Vol. 15, January 2014.

[2] Tripathi et al. *Study of Limb Darkening Effect and Rotation Period of Sun by using Solar Telescope.* Journal of Scientific Research. Vol. 64, Issue 1, January 2020.

6 Appendices

6.1 Latitudinal and Longitudinal Coordinates

$$
\begin{aligned}
r\ :\ \langle \rho\cos\varphi\cos\theta\,,\,\rho\cos\varphi\sin\theta\,,\,\rho\sin\varphi\rangle \implies \begin{cases} r_\theta = \langle -\rho\cos\varphi\sin\theta\,,\,\rho\cos\varphi\cos\theta\,,\,0\rangle \\ r_\varphi = \langle -\rho\sin\varphi\cos\theta\,,\,-\rho\sin\varphi\sin\theta\,,\,\rho\cos\varphi\rangle \end{cases} \\
\implies \quad r_\theta\times r_\varphi &= \langle \rho^2\cos^2\varphi\cos\theta\,,\,\rho^2\cos^2\varphi\sin\theta\,,\,\rho^2\cos\varphi\sin\varphi\rangle \\
\implies \quad |r_\theta\times r_\varphi|^2 &= \rho^4\cos^4\varphi\cos^2\theta+\rho^4\cos^4\varphi\sin^2\theta+\rho^4\cos^2\varphi\sin^2\varphi \\
&= \rho^4\cos^4\varphi+\rho^4\cos^2\varphi\sin^2\varphi \\
&= \rho^4\cos^2\varphi
\end{aligned}
\tag{6.1.1}
$$

$$
\rho^2\int_0^{2\pi}\int_{\varphi_1}^{\varphi_2}\cos\varphi\ d\theta d\varphi = 2\pi\rho^2(\sin\varphi_2 - \sin\varphi_1)
\tag{6.1.2}
$$

6.2 Limb Darkening Normalization

$$
\begin{aligned}
\int_{-\pi/2}^{\pi/2}\sum_{i=0}^{3} a_i\cos^i\theta\ d\theta &= \pi a_0+2(a_1+a_3)+\frac{a_2}{2}\int_{-\pi/2}^{\pi/2} 1+\cos 2\theta\ d\theta - a_3\int_{-\pi/2}^{\pi/2}\cos\theta\sin^2\theta\ d\theta \\
&= \pi(a_0+\frac{a_2}{2})+2(a_1+a_3)-2\frac{a_3}{3} \\
&= \pi a_0+2a_1+\frac{\pi}{2}a_2 - \frac{4}{3}a_3
\end{aligned}
\tag{6.2.1}
$$

6.3 Planet-Shade Projection To The Star

$$
\begin{aligned}
r_\alpha^2 &= |(\mathbf{v}_\beta+t\mathbf{v}_{\beta\gamma})-\mathbf{c}_\alpha|^2 \\
&= |(\mathbf{v}_\beta-\mathbf{c}_\alpha)+t\mathbf{v}_{\beta\gamma}|^2 \\
&= |\mathbf{v}_\beta-\mathbf{c}_\alpha|^2+2t[\mathbf{v}_{\beta\gamma}\cdot(\mathbf{v}_\beta-\mathbf{c}_\alpha)]+t^2|\mathbf{v}_{\beta\gamma}|^2
\end{aligned}
\tag{6.3.1}
$$

6.4 Spherical Coordinate Transformation Bounds

$$
\begin{aligned}
\frac{\mathbf{a}\cdot\mathbf{b}}{ab} - \frac{\mathbf{a}'\cdot\mathbf{b}'}{a'b'} &= \frac{\mathbf{a}\cdot\mathbf{b} - \mathbf{a}'\cdot\mathbf{b}'}{ab} + \mathbf{a}'\cdot\mathbf{b}'\left(\frac{1}{ab} - \frac{1}{a'b'}\right) \\
&= \frac{(\mathbf{a} - \mathbf{a}')\cdot\mathbf{b} + \mathbf{a}'\cdot(\mathbf{b} - \mathbf{b}')}{ab} + \mathbf{a}'\cdot\mathbf{b}'\frac{(a'-a)b' + a(b'-b)}{aba'b'} \\
&\leq \delta\frac{a + b + a' + b'}{ab} \\
&\leq 2\delta\frac{a + b + \delta}{ab}
\end{aligned}
\tag{6.4.1}
$$

$$
\frac{d}{dx}\cos^{-1}x = \frac{-1}{(1-x^2)^{1/2}} \qquad x \in (-1,1) \tag{6.4.2}
$$